

# YUPU YAO

yupuyao.github.io

(+86) 18181918616 ◇ yypseek123@gmail.com

## EDUCATION

---

### University of Electronic Science and Technology of China

B.Eng. in Communication Engineering

*Sept 2022 - Expected June 2026*

### University of Glasgow

B.Eng. in Electronics and Electrical Engineering with Communications  
2026

*Sept 2022 - Expected June*

## WORK EXPERIENCE

---

### Infinigence AI

*Aug 2025 - Mar 2026*

*Inference Framework Developer Intern*

*Shanghai, China*

- Led the development of an SGLang-based and vllm-based heterogeneous inference framework, architecting an efficient GPU-FPGA scheduler to resolve distributed Attention and MoE bottlenecks via computation-communication overlap.
- Implemented Prefill/Decode phase separation to mitigate mutual interference between processing stages and enhance overall throughput.
- Successfully deployed and optimized large-scale models (DeepSeek 671B, Qwen 235B) supporting FP8/INT4 precision on distributed 8xH20 and 8xA100 clusters.

## RESEARCH EXPERIENCE

---

### Deconstructing Attention Mechanisms via Implicit Hilbert Ladder

*Feb 2024 - June 2024*

- Proposed and implemented the Implicit Hilbert Ladder (IHL) framework, modeling Multi-Head Self-Attention (MSA) from a Reproducing Kernel Hilbert Space (RKHS) perspective to provide a new theoretical understanding of Transformers.
- Developed a novel mechanism, Hilbertian Attention Cascade (HAC), replacing self-attention with kernel mappings in RKHS. Achieved a >5% accuracy increase in early-stage training on CIFAR-10, with 16% fewer parameters and a 12.3% reduction in GFLOPs.
- Validated the theoretical advantages of IHL via Rademacher complexity generalization bounds, Neural Tangent Kernel (NTK) convergence proofs, and gradient flow analysis, establishing a theoretically-grounded attention alternative

### Resource Allocation for Semantic-Aware Communication Networks

*Aug 2022 - Feb 2024*

- Deployed and trained a Deep Q-Network (DQN)-based intelligent resource allocation algorithm, improving average user Quality of Experience (QoE) by 23.5% and system convergence speed by 18% in simulated wireless networks.
- Proposed a distributed two-tier control strategy integrating local DQNs with a global coordination module, reducing training communication overhead by 28.6% while maintaining near-optimal global performance.
- Effectively addressed the joint task of beamforming, power control, and semantic resource allocation, demonstrating a 19% improvement in average policy robustness under multi-task, high-dimensional state spaces.

### VQ-GAN based Semantic Communication System

*Dec 2022 - Mar 2023*

- Proposed and implemented a novel end-to-end semantic communication architecture, introducing a VQ-GAN encoding module to replace traditional source-channel separation and achieve joint optimization of compression and representation.
- Improved semantic reconstruction accuracy by 21.3% and bandwidth efficiency by 35.7% compared to conventional systems.

## PUBLICATIONS AND MANUSCRIPTS

---

- [1] **Yupu Yao** . Generalize neural network through smooth hypothesis function. *ICLR 2024(Tiny)*
- [2] **Yupu Yao**, Bowen Yang. GeoPE: A Unified Geometric Positional Embedding for Structured Tensors. *In Preprint*
- [3] **Yupu Yao**, Shangqi Deng, Zihan Cao, Harry Zhang, Liang-jian Deng. APLA: Additional perturbation for latent noise with adversarial training enables consistency. *In Preprint*
- [4] **Yupu Yao**, Jin Qi. Dynamic Deconvolution Learning: Towards Interpretable Neural Signal Decomposition *Undergraduate Thesis*

## AWARDS AND SCHOLARSHIPS

---

**Provincial Gold Medal, The 9th China "Internet+" College Students' Innovation Competition** *Aug 2023*  
 Developed and deployed advanced remote sensing algorithms for applications in agriculture, environmental monitoring, disaster prediction, and defense.

## SERVICE

---

**Conference Reviewer**  
 NeurIPS *2026*

## TECHNICAL SKILLS

---

- **ML/DL Frameworks:** Pytorch, Tensorflow, JAX, Triton
- **Training Frameworks:** DeepSpeed
- **Inference Frameworks:** SGLang, Vllm